

Simulating Survival Data

2026-04-17

Introduction

Simulating censored time-to-event data is essential for three classes of work: power and sample-size calculations for trials with non-standard designs, validation of new statistical methods (does the estimator recover the truth?), and pedagogy. The inverse-hazard method generates event times from any specified hazard function and lets you build realistic datasets that match a target survival pattern, censoring distribution, and covariate-effect structure.

Prerequisites

A working understanding of hazard and cumulative hazard functions, the inverse-CDF method for random-number generation, and the relationship between proportional-hazards models and exponentiated covariate effects.

Theory

For a hazard $h(t \mid x)$ with cumulative hazard $H(t \mid x)$ and a uniform variate $U \sim \text{Uniform}(0, 1)$,

$$T = H^{-1}(-\log U \mid x)$$

has the target distribution. Under proportional hazards $H(t \mid x) = H_0(t) \exp(x^\top \beta)$, so

$$T_i = H_0^{-1}(-\log U_i / \exp(x_i^\top \beta)).$$

Censoring is added by drawing an independent censoring time C_i from a chosen distribution (administrative, exponential, or empirical from a real cohort) and recording $\min(T_i, C_i)$ as the observation time and $\mathbf{1}\{T_i \leq C_i\}$ as the event indicator.

Assumptions

The specified baseline hazard and covariate-effect structure are the truth being simulated; the censoring distribution is independent of T_i unless the simulation explicitly handles informative censoring.

R Implementation

```
library(simsurv)

set.seed(2026)
n <- 300
x <- data.frame(id = 1:n, arm = factor(rep(c("A", "B"), each = n/2)))
x$arm_num <- as.numeric(x$arm) - 1

# Weibull baseline: lambda = 0.1, gamma = 1.2
sim <- simsurv(lambdas = 0.1, gammas = 1.2,
```

```

      x = x, betas = c(arm_num = -0.5),
      maxt = 10)

# Merge with covariates
d <- merge(sim, x, by = "id")
head(d)

# Kaplan-Meier
library(survival)
fit_km <- survfit(Surv(eventtime, status) ~ arm, data = d)
plot(fit_km)

```

Output & Results

`simsurv()` returns a tibble with one row per subject containing the simulated event time, the censoring indicator (here taken at `maxt = 10`), and the subject id. Merging back with the covariate frame gives a complete simulated dataset ready for analysis. The Kaplan-Meier plot shows the two arms separating with the designed effect size.

Interpretation

A reporting sentence: “Simulated Weibull data with shape 1.2, baseline scale 0.1, and a treatment hazard ratio of $\exp(-0.5) = 0.61$; a Cox model on the simulated data recovered an HR of 0.59 (95 % CI 0.46 to 0.76), confirming the simulation is correctly calibrated.” Always confirm that the analysis pipeline recovers the simulation parameters before using the simulator for power calculations.

Practical Tips

- `simsurv` supports Weibull and Gompertz baselines plus user-defined hazard functions; for arbitrary parametric shapes, supply `hazard` or `loghazard` directly.
- For competing risks, simulate each cause’s event time independently and record the cause that occurred first; the resulting data have the correct cumulative incidence functions.
- Match the censoring distribution to the real-study target: administrative censoring at study end, plus loss-to-follow-up exponentials with rates calibrated to historical drop-out.
- For power calculations, run thousands of simulated datasets, fit the planned model on each, and record the proportion of times the test rejects at the chosen alpha; this is the standard Monte Carlo power estimator.
- For time-varying covariates, simulate via piecewise-constant hazards across pre-specified intervals; `simsurv` supports `tdefunction` for time-dependent covariate effects.
- For validation of new methods, re-simulate under several scenarios (PH violation, heavy censoring, sparse events) and check robustness; a method that works only on its design assumptions is brittle.

R Packages Used

`simsurv` for the canonical inverse-hazard simulator with Weibull, Gompertz, and user-defined baselines; `flexsurv` for `rflexsurv()` simulation from fitted flexible models; `coxed` for Cox-model-based simulations; `survSim` for additional simulation extensions including frailty and time-dependent effects.

For Reviewers

What to look for in a paper using this method.

- Common misapplications.
- Diagnostics that should be reported but often aren’t.

- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Survival primer: KM, log-rank, Cox PH
- Time-varying covariates and immortal-time bias
- Competing risks and multistate models
- Survival ML (random survival forest, DeepSurv)
- Time-dependent Brier, IPA, external validation